

Quiz 8: The Exponential–Gaussian Connection

1. (2 points) **Simulating a geometric random variable via an exponential one.**

Let $E \sim \text{Exp}(1)$. Fix $p \in (0, 1)$ and define

$$c := -\log(1 - p) > 0, \quad G := \left\lfloor \frac{E}{c} \right\rfloor + 1.$$

Show that for $k \geq 1$

$$\mathbb{P}(G = k) = \mathbb{P}((k - 1)c \leq E < kc).$$

Use this to conclude that $G \sim \text{Geom}(p)$.

2. (2 points) **Loss of concentration in the exponential distribution.**

Let $T_\lambda \sim \text{Exp}(\lambda)$ where $\lambda > 0$. Fix a constant $c > 0$.

(a) Show that

$$\mathbb{P}\left(\left|T_\lambda - \frac{1}{\lambda}\right| \leq c\right) = e^{-1+\lambda c} - e^{-1-\lambda c}.$$

(b) Show that

$$\mathbb{P}\left(\left|T_\lambda - \frac{1}{\lambda}\right| \leq c\right) \rightarrow 0 \quad \text{as } \lambda \downarrow 0,$$

and explain in words what this says about the concentration of T_λ around its mean.

3. (2 points) **Geometry of the Gaussian density.**

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ where $\mu \in \mathbb{R}$ and $\sigma > 0$. Its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

(a) Show that the maximum of $f(x)$ occurs at $x = \mu$.

(b) Show that the inflection points of $f(x)$ occur at $x = \mu \pm \sigma$.

(c) Determine the regions where the graph of $f(x)$ is concave up and concave down.

(d) Using the previous parts, sketch the graph of $f(x)$ and label the maximum, the inflection points, the regions where the graph is concave up and concave down, and the tail behavior.

(e) Explain briefly why the Gaussian density is often described as *bell-shaped*.

4. (2 points) **Gaussian concentration.**

Let $Z \sim \mathcal{N}(0, 1)$.

(a) Show that for every $s \in \mathbb{R}$, $\mathbb{E}[e^{sZ}] = e^{s^2/2}$.

- (b) Use Markov's inequality to prove that for every $t > 0$, $\mathbb{P}(Z \geq t) \leq e^{-t^2/2}$.
- (c) Deduce the two-sided bound $\mathbb{P}(|Z| \geq t) \leq 2e^{-t^2/2}$.

5. (2 points) **Box–Muller Method and the Exponential Distribution.**

Let $U_1, U_2 \sim \text{Unif}(0, 1)$ be independent and define

$$R := \sqrt{-2 \log U_1}, \quad \Theta := 2\pi U_2.$$

(a) Show that $R^2 = -2 \log U_1 \sim \text{Exp}(1/2)$.

(b) Let

$$X = R \cos \Theta, \quad Y = R \sin \Theta.$$

Show that the joint density of (X, Y) is

$$f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}, \quad (x, y) \in \mathbb{R}^2,$$

and conclude that X and Y are independent $\mathcal{N}(0, 1)$ random variables.